Sequences

Lecture and worked examples

A sequence (u_n) is **arithmetic** when there is a real number r such that, for all n, $u_{n+1} = u_n + r$. The number r is the common difference. If the first term is u_0, then $u_n = u_0 + nr$. If the first term is u_1, then $u_n = u_1 + (n-1)r$.

Core Course Notes

To model an additive situation: 1) define u_n and state the unit; 2) identify the first term; 3) express the constant increase or decrease as the common difference r; 4) write the general term; 5) answer with a sentence in context.

A shop sells 480 units in the first month and then 25 more units each month. With $u_1 = 480$ and $r = 25$, we get $u_n = 480 + 25(n - 1)$. Thus $u_{12} = 755$: the shop expects to sell 755 units in the twelfth month.

In $u_3 + u_4 + \cdots + u_{12}$, there are $12 - 3 + 1 = 10$ terms, not 9 or 12.

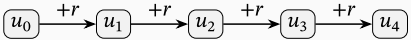
A sequence (u_n) is **geometric** when there is a real number q such that $u_{n+1} = qu_n$. If the first term is u_0, then $u_n = u_0 q^n$. If the first term is u_1, then $u_n = u_1 q^{n-1}$.

Arithmetic sequences: recognising and modelling

- Recognise an additive change.
- Find the common difference.
- Define a sequence in context.

Arithmetic sequences: recognising and modelling

A club has 320 members. It expects to gain 18 more each year. Let u_n be the number of members after n years. Calculate u_0, u_1, u_2. Which operation takes you from one term to the next? Write a relation between u_{n+1} and u_n, then predict u_5.



Arithmetic sequences: recognising and modelling

A sequence is arithmetic when $u_{n+1} - u_n = r$ is constant. In a real-life situation, a constant absolute increase leads to an arithmetic sequence.

Arithmetic sequences: recognising and modelling

Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Arithmetic sequences: recognising and modelling

1 h session: 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.

2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.

Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

Arithmetic sequences: the general term

- Move from the recurrence relation to the general term.
- Calculate a distant term.
- Interpret the index.

Arithmetic sequences: the general term

A shop has 1,250 active loyalty cards at index 0 and gains 75 per month. Without calculating all the terms, find the number of cards at index n, then at index 18. Explain the role of 18 in the calculation.

Arithmetic sequences: the general term

If (u_n) is arithmetic with common difference r, then $u_n = u_0 + nr$ or $u_n = u_1 + (n-1)r$, depending on the index of the first term.

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Sum of the terms of an arithmetic sequence

- Count the terms of a sum.
- Apply the formula.
- Check the order of magnitude.

Sum of the terms of an arithmetic sequence

A company pays a monthly bonus: 45 euros in the first month, then 5 euros more each month. Calculate the total paid over the first twelve months in three ways: explicit sum, pairing first and last terms, formula.

Sum of the terms of an arithmetic sequence

A sum of arithmetic terms equals number of terms $\times \dfrac{\text{first} + \text{last}}{2}$. From u_p to u_q, there are $q - p + 1$ terms.

Sum of the terms of an arithmetic sequence

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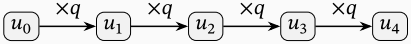
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Geometric sequences: recognising and modelling

- Recognise a multiplicative change.
- Convert a rate into a multiplier.
- Compare arithmetic and geometric sequences.

Geometric sequences: recognising and modelling

An online shop increases its number of visitors by 6% per month, starting from 8,000 visitors. Calculate the first three values. Compare the differences, then the quotients. Identify the type of sequence.



Geometric sequences: recognising and modelling

A sequence is geometric when $\frac{u_{n+1}}{u_n} = q$ is constant (non-zero terms). A constant relative change leads to a geometric sequence.

Geometric sequences: recognising and modelling

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Geometric sequences: recognising and modelling

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Geometric sequences: general term and sum

- Write the general term.
- Calculate a geometric sum.
- Interpret a cumulative change.

Geometric sequences: general term and sum

A capital of 250,000 Ar is invested at 4% per year, compound interest. Write the capital C_n after n years. Calculate C_6, then the sum $C_0 + \cdots + C_6$, and explain what this last sum represents mathematically and why it is not the final capital.

Geometric sequences: general term and sum

If (u_n) is geometric with common ratio q, $u_n = u_0 q^n$ ou $u_n = u_1 q^{n-1}$. For $q \neq 1$, $u_0 + \cdots + u_n = u_0 \dfrac{1 - q^{n+1}}{1 - q}$.

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Review and formative assessment

- Choose the right model.
- Use the formulas.
- Analyse one's own mistakes.

Review and formative assessment

Sort eight situations into three categories: arithmetic, geometric, neither. For each situation, underline the word or piece of data that justifies the choice, then choose the appropriate formula.

1. A rent of 600,000 Ar increases by 20,000 Ar each year.
2. A town with 5,000 inhabitants sees its population grow by 3% per year.
3. A club membership costs 15 euros to join, then 2 euros per session: let u_n be the total cost after n sessions.
4. A vehicle loses 12% of its value each year.
5. The temperatures recorded at noon on four days in a row are 21, 23, 22 and 25 degrees.
6. A stock of 900 items decreases by 45 items each week.
7. The number of members of a club, year by year, is 40, 44, 49 and 55.

8. A capital of 2,000 euros is invested at 2.5% per year, compound interest.

Review and formative assessment

Before calculating, identify: type of sequence, first index, common difference or ratio, and the term or sum asked for. The conclusion states the unit and the index.

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